
Determinacy and Large-Scale Solutions in the Sequence Space

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Roadmap for determinacy and existence

- ❖ Want a sequence-space criterion for determinacy and existence
- ❖ To get there: obtain a **structure theorem** for sequence-space Jacobians
- ❖ When het-agent model is *stationary*, Jacobians are *quasi-Toeplitz*:

$$\mathbf{J} = T(\mathbf{j}) + \mathbf{E}$$

i.e. sum of Toeplitz operator $T(\mathbf{j})$ and compact operator $\mathbf{E}$ on ℓ^2

- ❖ Will exploit this structure in many ways, but start with:
 - ❖ “**Winding number**” criterion on $\mathbf{j}$ for **determinacy & existence**

Toeplitz and quasi-Toeplitz operators

What is a Toeplitz operator?

- ❖ We'll work with **semi-infinite** Toeplitz matrices $T(\mathbf{j})$ with constant diagonals $\{j_s\}_{s=-\infty}^{\infty}$
- ❖ Assuming $\sum_s |j_s| < \infty$, these induce bounded operators on ℓ^2
- ❖ Can define series $j(z) \equiv \sum_{s=-\infty}^{\infty} j_s z^s$, sometimes called "symbol" of $T(\mathbf{j})$

$$T(\mathbf{j}) = \begin{pmatrix} j_0 & j_{-1} & j_{-2} & \ddots \\ j_1 & j_0 & j_{-1} & \ddots \\ j_2 & j_1 & j_0 & \ddots \\ \ddots & \ddots & \ddots & \ddots \end{pmatrix}$$

Examples: lag and lead operators

$$\begin{pmatrix} 0 & 0 & 0 & \dots \\ 1 & 0 & 0 & \ddots \\ 0 & 1 & 0 & \ddots \\ \vdots & \ddots & \ddots & \ddots \end{pmatrix}$$

Lag operator $\mathbf{L} : (x_0, x_1, x_2, \dots) \mapsto (0, x_0, x_1, \dots)$

[Injective but not surjective (range missing 1 dimension)]

$$\begin{pmatrix} 0 & 1 & 0 & \dots \\ 0 & 0 & 1 & \ddots \\ 0 & 0 & 0 & \ddots \\ \vdots & \ddots & \ddots & \ddots \end{pmatrix}$$

Lead operator $\mathbf{F} : (x_0, x_1, x_2, \dots) \mapsto (x_1, x_2, x_3, \dots)$

[Surjective but not injective (null-space has 1 dimension)]

[= taking one-period-ahead expectations given MIT shock]

More complex example: reset prices given marginal cost

- ❖ Log-linearizing standard Calvo model, get:

$$p_t^* = (1 - \beta\theta) \cdot \sum_{s=0}^{\infty} (\beta\theta)^s \cdot \mathbb{E}_t MC_{t+s}$$

- ❖ For MIT shock, mapping from $\{MC_t\}_{t=0}^{\infty}$ to $\{p_t^*\}_{t=0}^{\infty}$ is Toeplitz, equal to $(1 - \beta\theta)$ times matrix on right

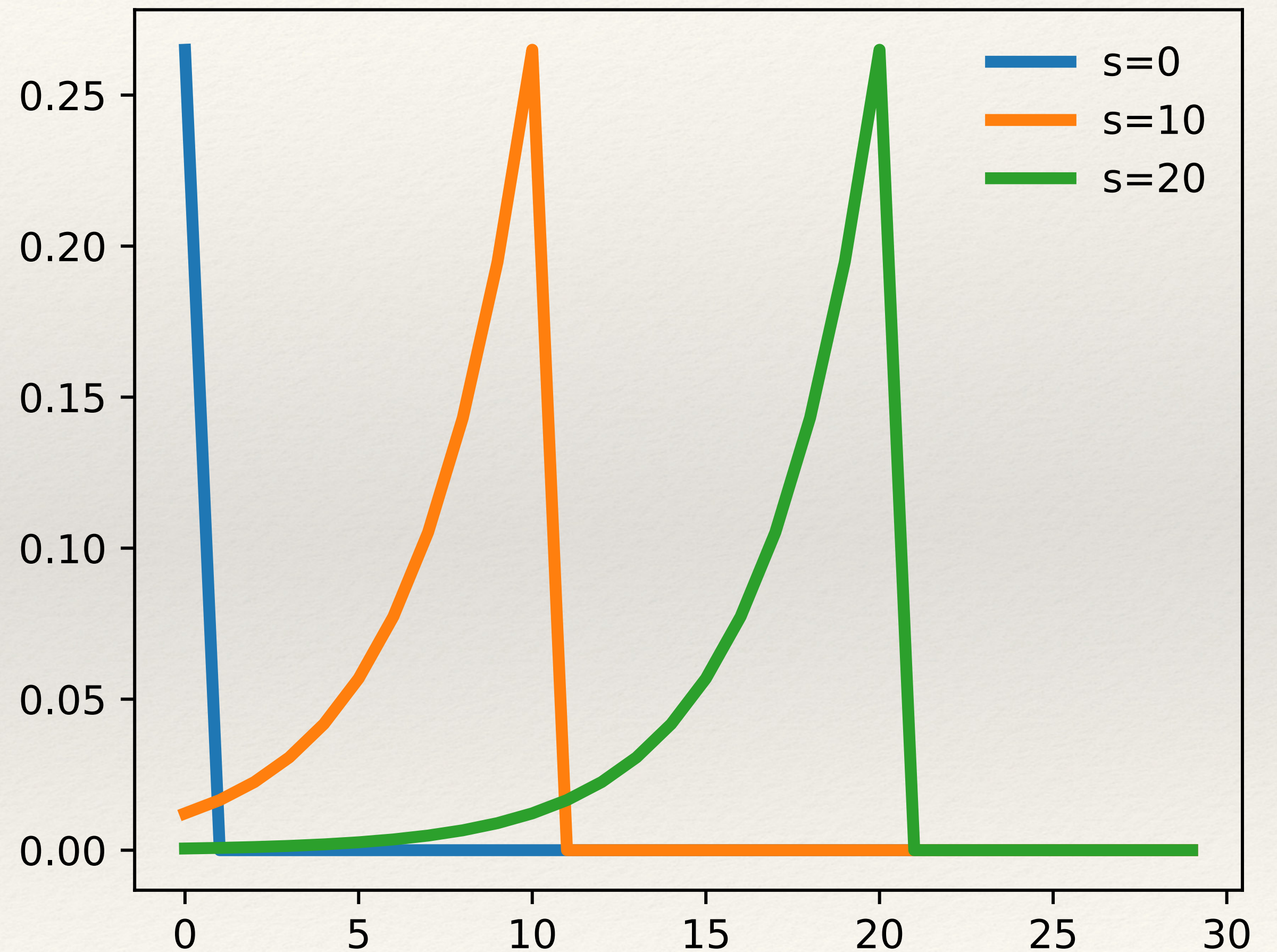
[forward-looking $\rightarrow$ upper triangular]

$$\begin{pmatrix} 1 & \beta\theta & (\beta\theta)^2 & \dots \\ 0 & 1 & \beta\theta & \dots \\ 0 & 0 & 1 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Columns of Toeplitz matrix: costs to reset prices

$$(1 - \beta\theta) \times \begin{pmatrix} 1 & \beta\theta & (\beta\theta)^2 & & \\ 0 & 1 & \beta\theta & \ddots & \\ 0 & 0 & 1 & \ddots & \\ & \ddots & & \ddots & \ddots \end{pmatrix}$$

$$(\beta = 0.98, \theta = 0.75)$$



Aggregate prices given reset prices

- ❖ Again log-linearizing standard Calvo model, we get:

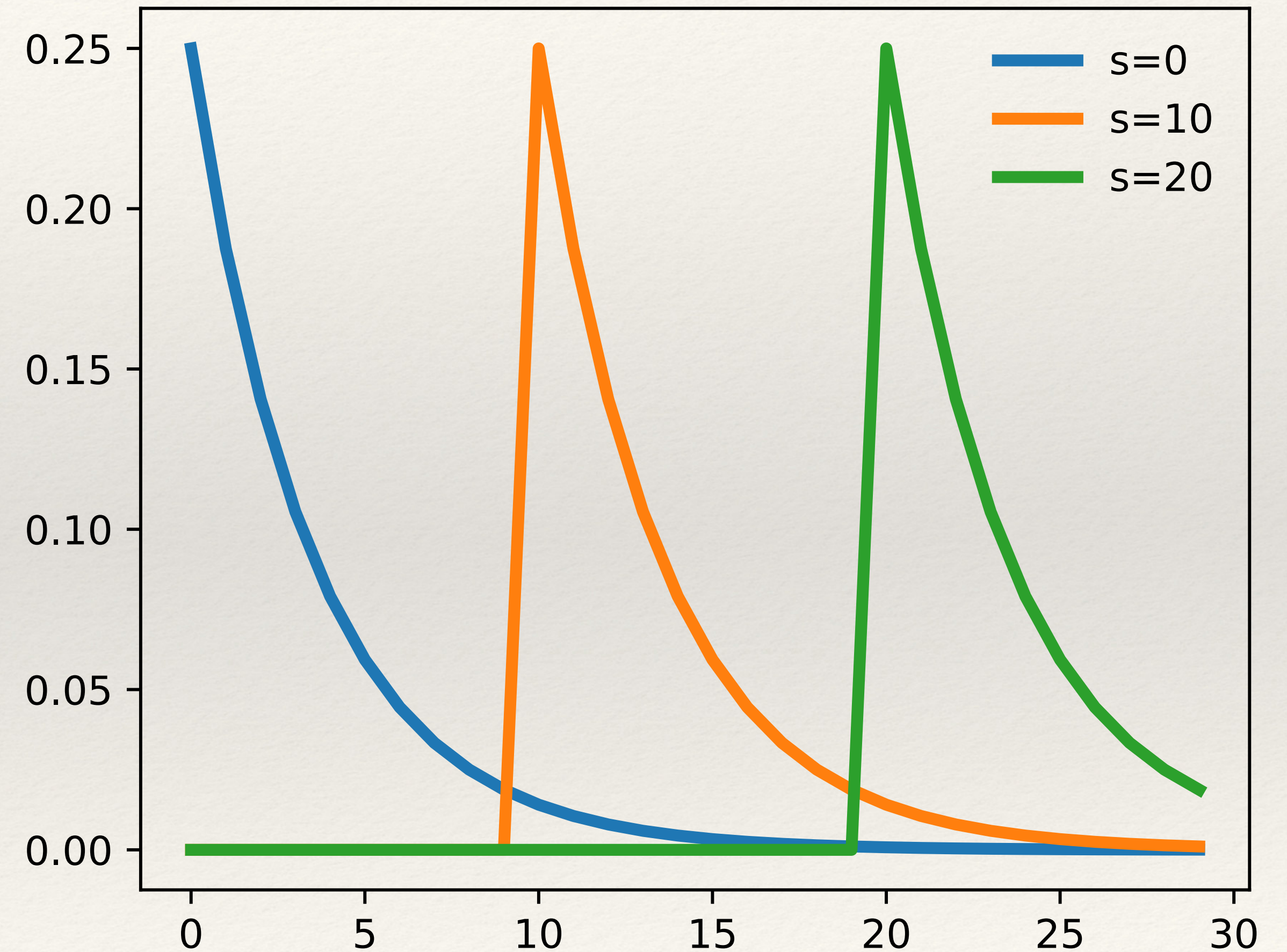
$$p_t = (1 - \theta) \cdot \sum_{s=0}^{\infty} \theta^s \cdot p_{t-s}^*$$

- ❖ Like before, mapping from $\{p_t^*\}_{t=0}^{\infty}$ to $\{p_t\}_{t=0}^{\infty}$ is Toeplitz, equal to $(1 - \theta)$ times matrix on right
[backward-looking $\rightarrow$ lower triangular]

$$\begin{pmatrix} 1 & 0 & 0 & \ddots \\ \theta & 1 & 0 & \ddots \\ \theta^2 & \theta & 1 & \ddots \\ \vdots & \ddots & \ddots & \ddots \end{pmatrix}$$

Columns of Toeplitz matrix: reset prices to agg prices

$$(1 - \theta) \times \begin{pmatrix} 1 & 0 & 0 & \ddots \\ \theta & 1 & 0 & \ddots \\ \theta^2 & \theta & 1 & \ddots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$



What if we want to compose Toeplitz operators?

- ❖ Would be nice to deal only with Toeplitz operators
- ❖ **Problem:** class of Toeplitz operators not closed under composition / multiplication
- ❖ Simple example: $\mathbf{FL} = \mathbf{I}$ (lead of lag is identity), but $\mathbf{LF} \neq \mathbf{I}$, and instead:

$$\mathbf{LF} = \begin{pmatrix} 0 & 0 & 0 & \cdots \\ 0 & 1 & 0 & \ddots \\ 0 & 0 & 1 & \ddots \\ \vdots & \ddots & \ddots & \ddots \end{pmatrix}$$

- ❖ So $(\mathbf{LF}) \cdot (x_0, x_1, x_2, \dots) = (0, x_1, x_2, \dots)$
- ❖ Interpretation for MIT shocks? "**Missing anticipation.**"
 - ❖ $\mathbb{E}_{t-1}x_t = x_t$ for $t > 0$, but $\mathbb{E}_{-1}x_0 = 0 \neq x_0$

But we **do** stay in larger quasi-Toeplitz class

- ❖ Define **quasi-Toeplitz** operators as Toeplitz operator plus compact operator **E**
[compact in ℓ^2 = can be uniformly well-approximated by finite rank]

$$\mathbf{J} = T(\mathbf{j}) + \mathbf{E}$$

- ❖ **Result:** product of any two Toeplitz operators $T(\mathbf{j}_1)$ and $T(\mathbf{j}_2)$ is quasi-Toeplitz like above, with $\mathbf{j} = \mathbf{j}_1 \cdot \mathbf{j}_2$ given by convolution [so $j(z) = j_1(z)j_2(z)$]

- ❖ Intuitively, why do we need **E**? “Missing anticipation”: taking lags of leads!
- ❖ Quasi-Toeplitz operators closed under multiplication!
[multiplying Toeplitz parts $\rightarrow$ quasi-Toeplitz, multiplying any bounded by compact $\rightarrow$ compact]

Compose to get map **J** from costs to aggregate prices

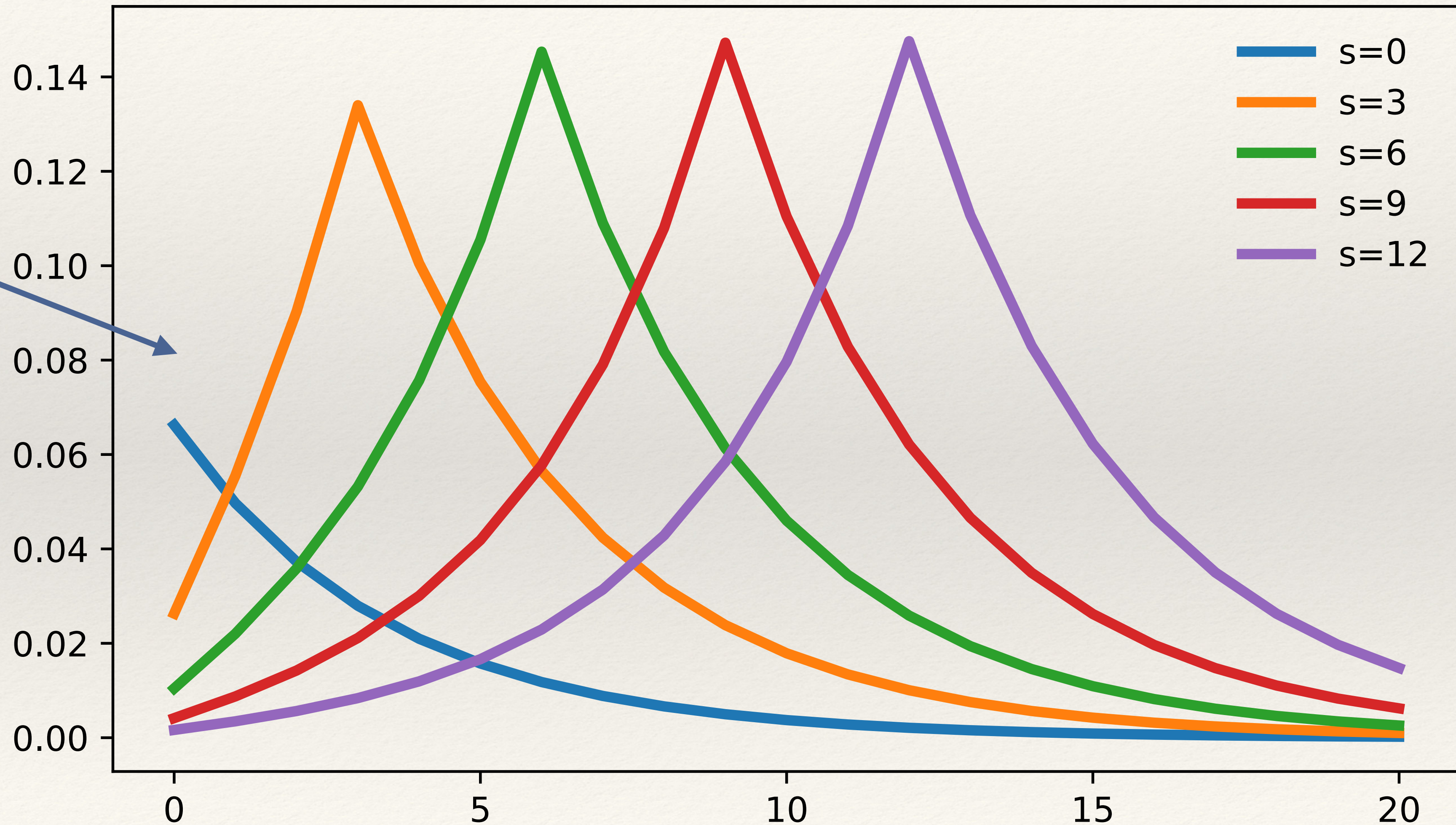
$$\begin{matrix} (1 - \theta) \times \\ \begin{pmatrix} 1 & 0 & 0 & \ddots \\ \theta & 1 & 0 & \ddots \\ \theta^2 & \theta & 1 & \ddots \\ \ddots & \ddots & \ddots & \ddots \end{pmatrix} \end{matrix} \times \begin{matrix} (1 - \beta\theta) \times \\ \begin{pmatrix} 1 & \beta\theta & (\beta\theta)^2 & \ddots \\ 0 & 1 & \beta\theta & \ddots \\ 0 & 0 & 1 & \ddots \\ \ddots & \ddots & \ddots & \ddots \end{pmatrix} \end{matrix}$$

(Lag of lead: we should expect a compact “correction” **E**!)

Visualizing columns of this $\mathbf{J}$

Eventually
converge to
Toeplitz pattern,
but “missing
anticipation”
leads to a limited
price response
near date 0.

Deviation $\mathbf{E}$ from
Toeplitz is
compact (and,
here, negative)



Structure theorem for heterogeneous-agent models

Structure theorem for heterogeneous-agent models

- ❖ Such models have some steady-state transition matrix Λ_{ss} , some backward mapping $\mathbf{v}_{t+1} \rightarrow \mathbf{v}_t$ on value function, with steady-state derivative $\mathbf{v}_v$
- ❖ Suppose we have “stationarity”:
 - ❖ Λ_{ss} has all eigenvalues but one strictly inside unit circle
 - ❖ $\mathbf{v}_v$ has eigenvalues strictly inside unit circle
- ❖ **Structure theorem:** if model is stationary, all sequence-space Jacobians are **quasi-Toeplitz** with

$$|E_{ts}| \leq K\Delta^{t+s}$$

for bound Δ on eigenvalues and some constant K

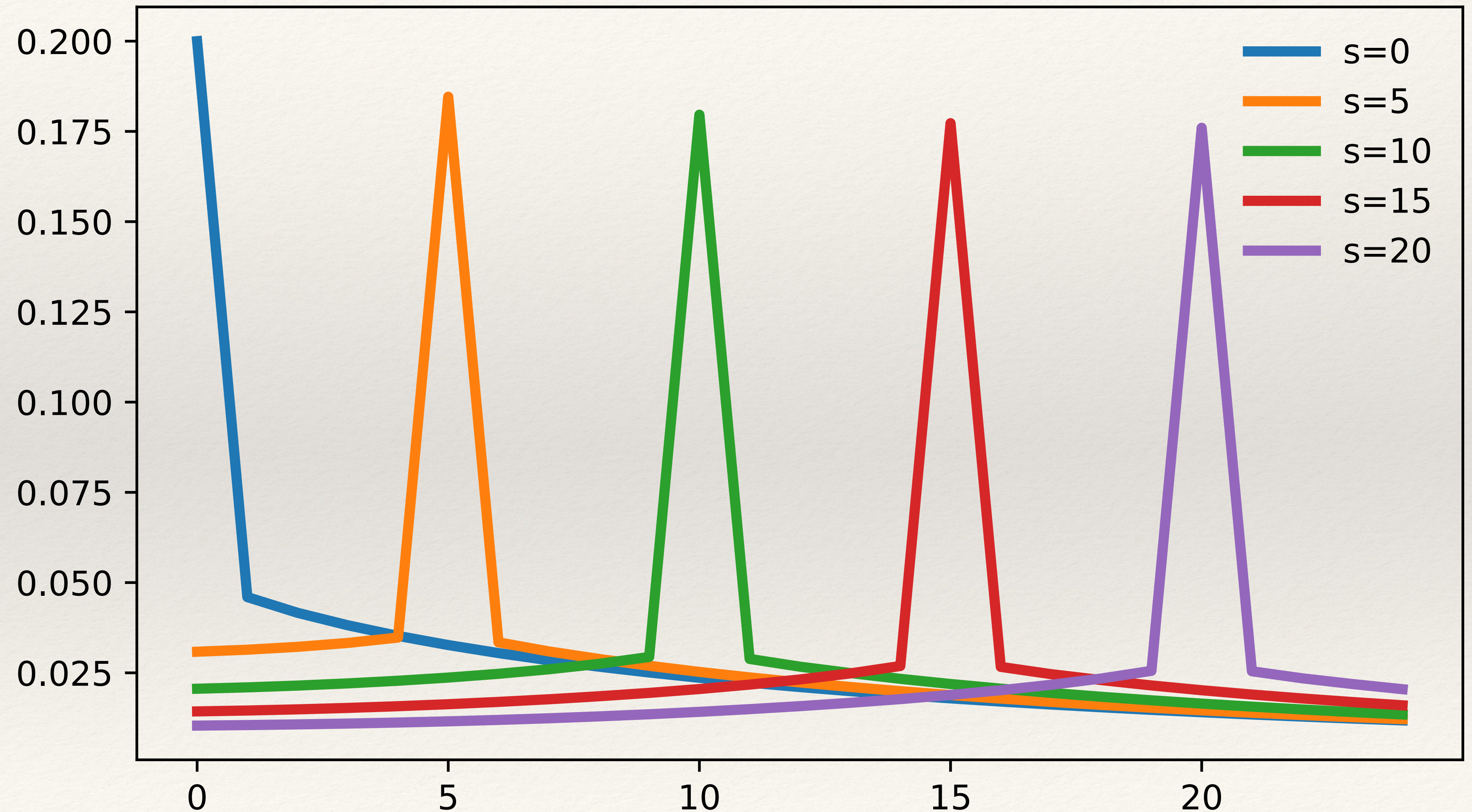
- ❖ Intuition: if effect of far future on value function eventually dies off...
 - ❖ ... and effect of distribution on future eventually dies off, “missing anticipation” dies off too

Example: intertemporal MPCs in a SIM model

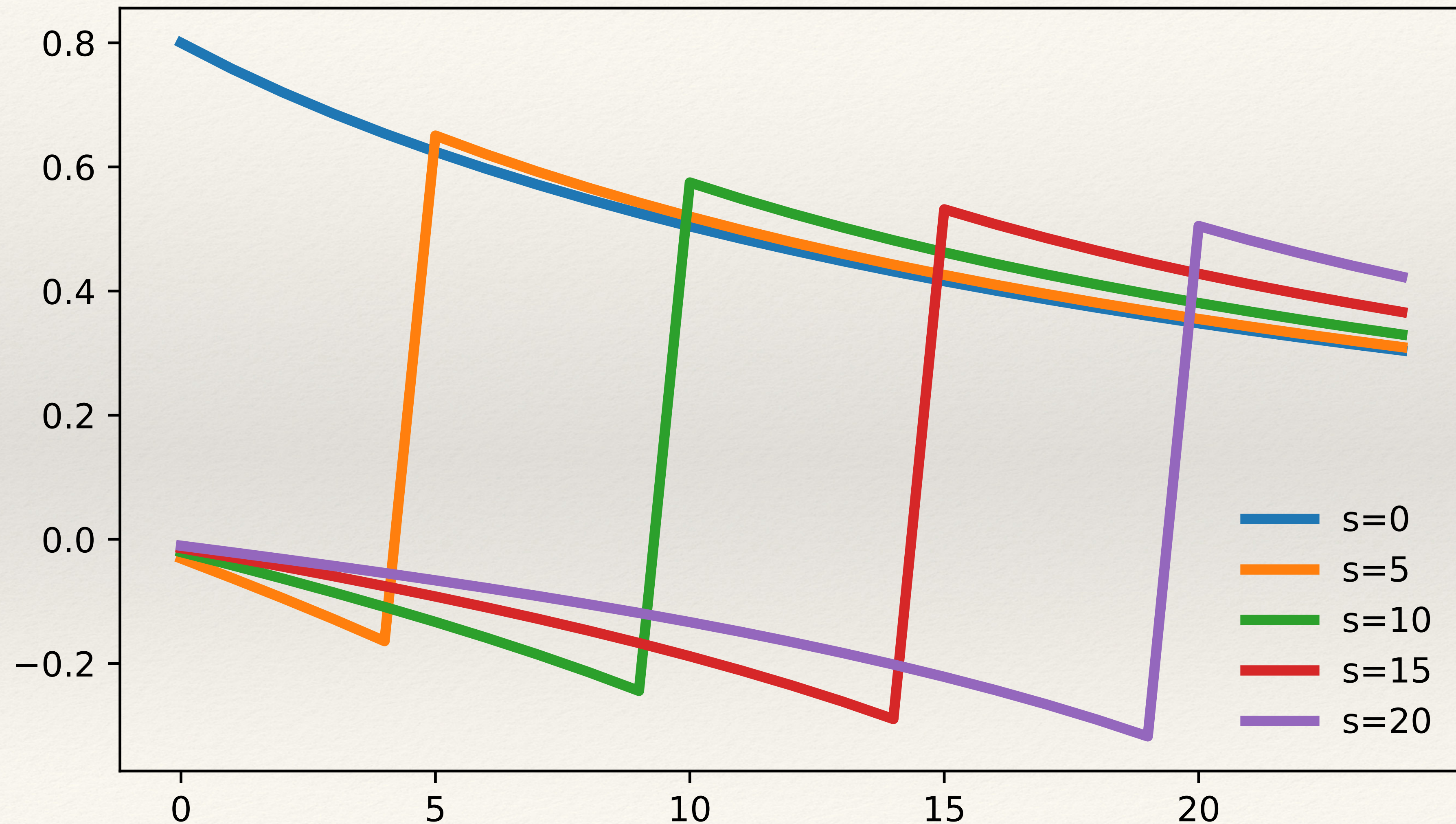
E is positive,
because not
having
anticipated
income means
you didn't spend
ahead of time,
leaving more to
spend now



(Here using our $A=100\%$
of GDP calibration from
earlier notebooks.)



Same, but Jacobian of assets vs. income



Taking stock

- ❖ We've shown that **quasi-Toeplitz** Jacobians naturally emerge:
 - ❖ as closure under multiplication of Toeplitz Jacobians (which themselves emerge from simple aggregate equations)
 - ❖ as Jacobians of stationary heterogeneous-agent problems
- ❖ Now will discuss **applications**:
 - ❖ **now**: for testing **determinacy** and **existence** of solutions
 - ❖ **later**: directly getting sequence-space solutions in “**truncation-free**” way, and solving **huge sequence-space systems**

Determinacy, existence, and inversion

Winding number of a Toeplitz operator

- ❖ **Winding number** $\text{wind}(j)$ is # of times symbol $j(z)$ rotates counterclockwise around 0 as z goes counterclockwise around the unit circle [see Onatski 2006]
- ❖ **Result:** can invert Toeplitz $T(\mathbf{j})$, resulting in quasi-Toeplitz, iff winding number 0
- ❖ For quasi-Toeplitz $T(\mathbf{j}) + \mathbf{E}$, this result is “generic” on open & dense set of $\mathbf{E}$:
 - ❖ If $\text{wind}(j) = 0$, $\mathbf{J}$ is generically **invertible**
 - ❖ If $\text{wind}(j) < 0$, $\mathbf{J}$ is **not injective (indeterminacy)**, but generically surjective
 - ❖ If $\text{wind}(j) > 0$, $\mathbf{J}$ is **not surjective (nonexistence)**, but generically injective
- ❖ [Example: lag operator has $j(z) = z$, winding number of 1, so not surjective.]

Application: uniqueness in Intertemporal Keynesian Cross model

- ❖ In Intertemporal Keynesian Cross model (Auclert Rognlie Straub 2024), equilibrium output $d\mathbf{Y}$ given taxes $d\mathbf{T}$ and bonds $d\mathbf{B}$ given by

$$\mathbf{A}(d\mathbf{Y} - d\mathbf{T}) = d\mathbf{B}$$

where $\mathbf{A}$ is Jacobian of household assets to post-tax income

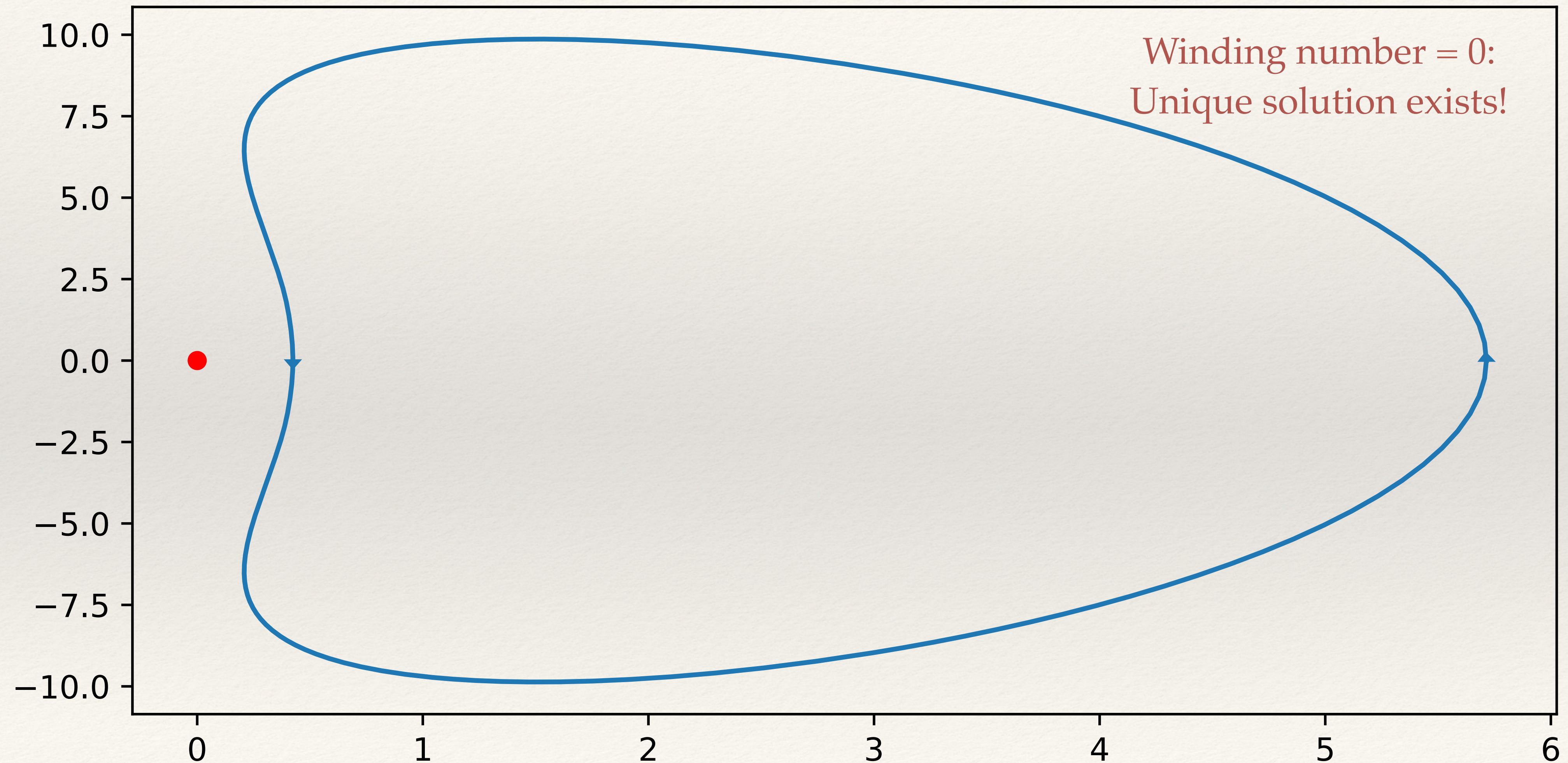
- ❖ $\mathbf{A}$ is quasi-Toeplitz, will generically be invertible with solution

$$d\mathbf{Y} = \mathbf{A}^{-1}d\mathbf{B} + d\mathbf{T}$$

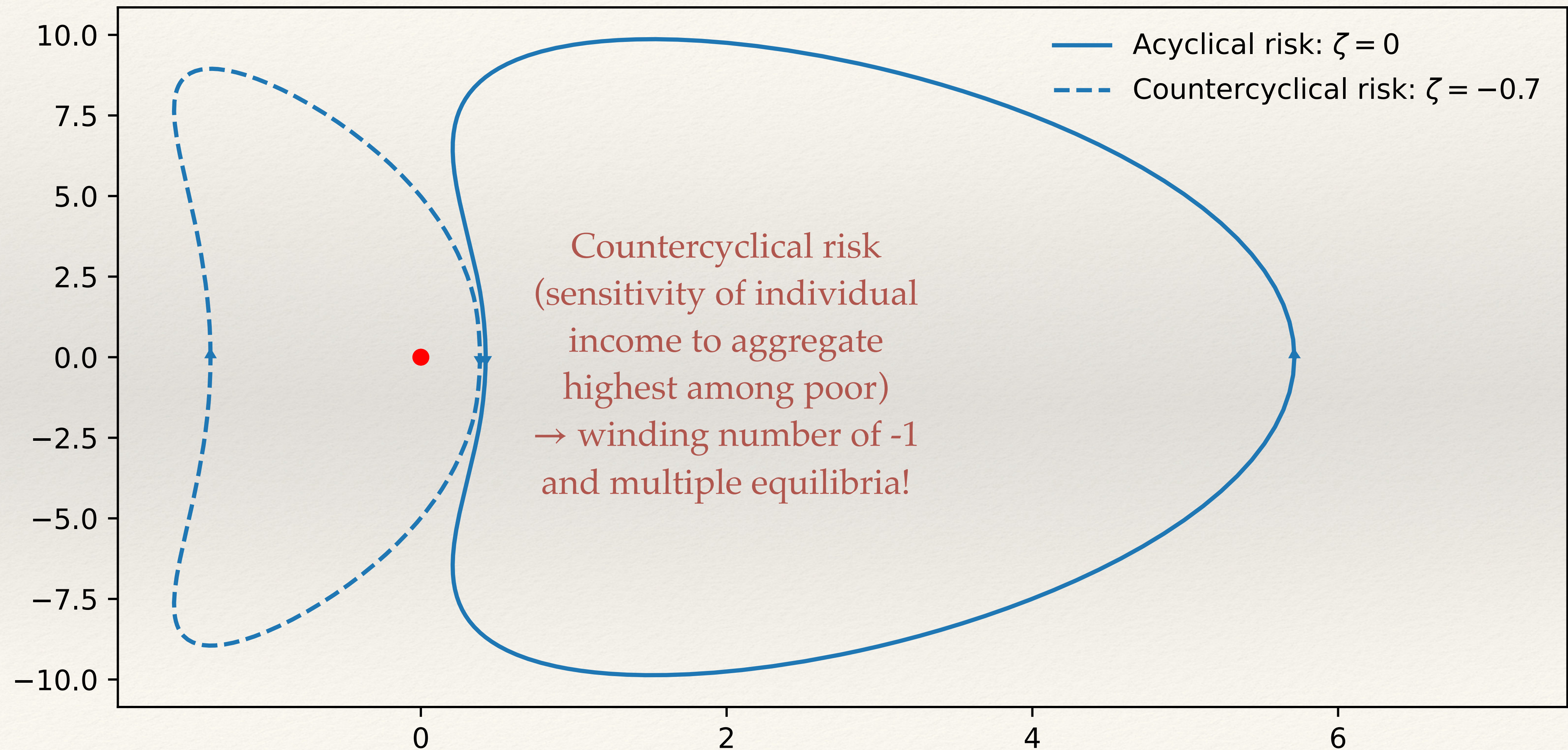
if winding number of $a(z)$ is zero

- ❖ Multiple equilibria $d\mathbf{Y}$ if winding number is negative
 - ❖ (Self-fulfilling booms or busts in output!)

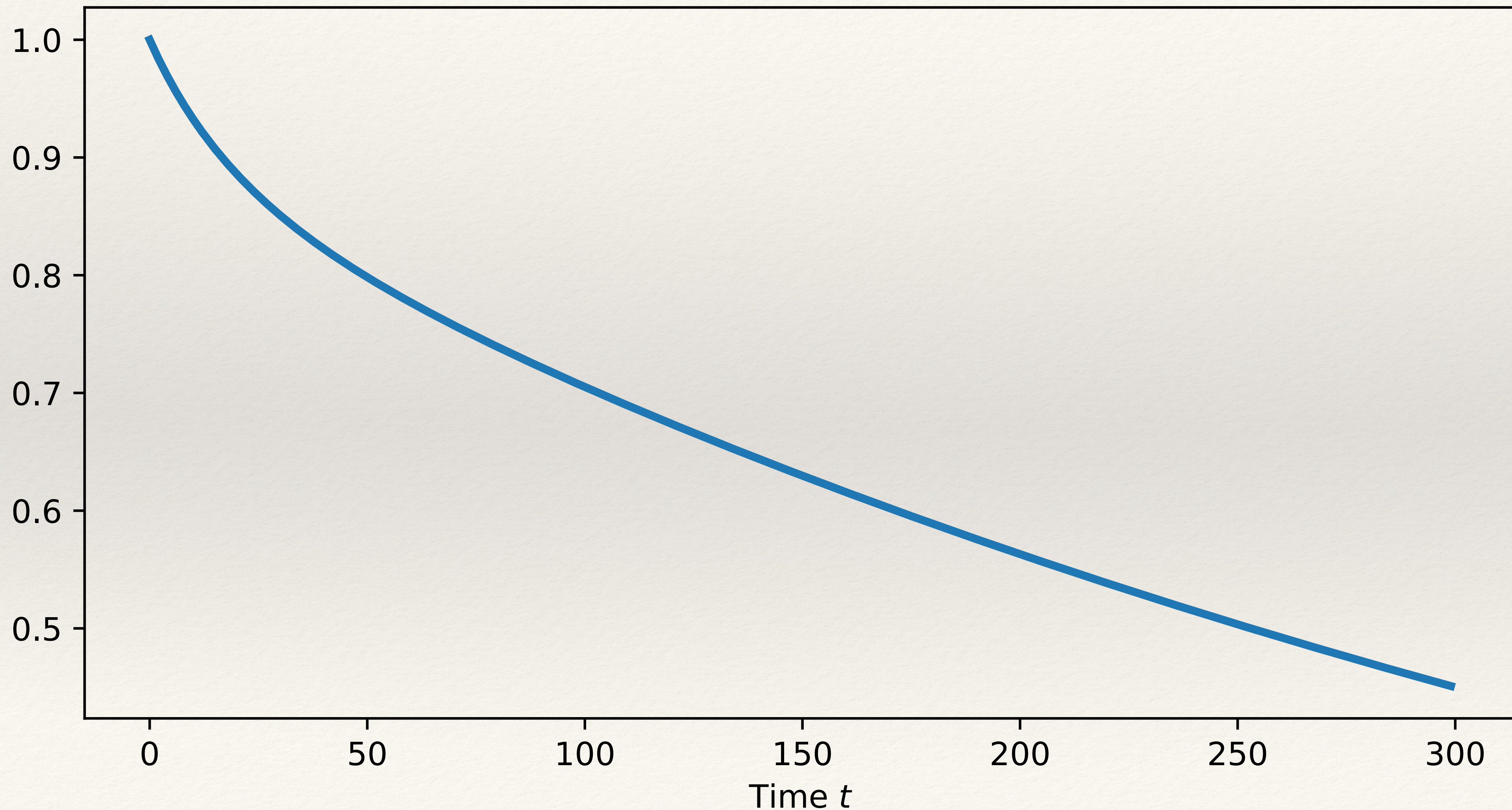
Winding number plot for $\mathbf{A}$ with a standard calibration



Case with countercyclical risk: **self-fulfilling boom**

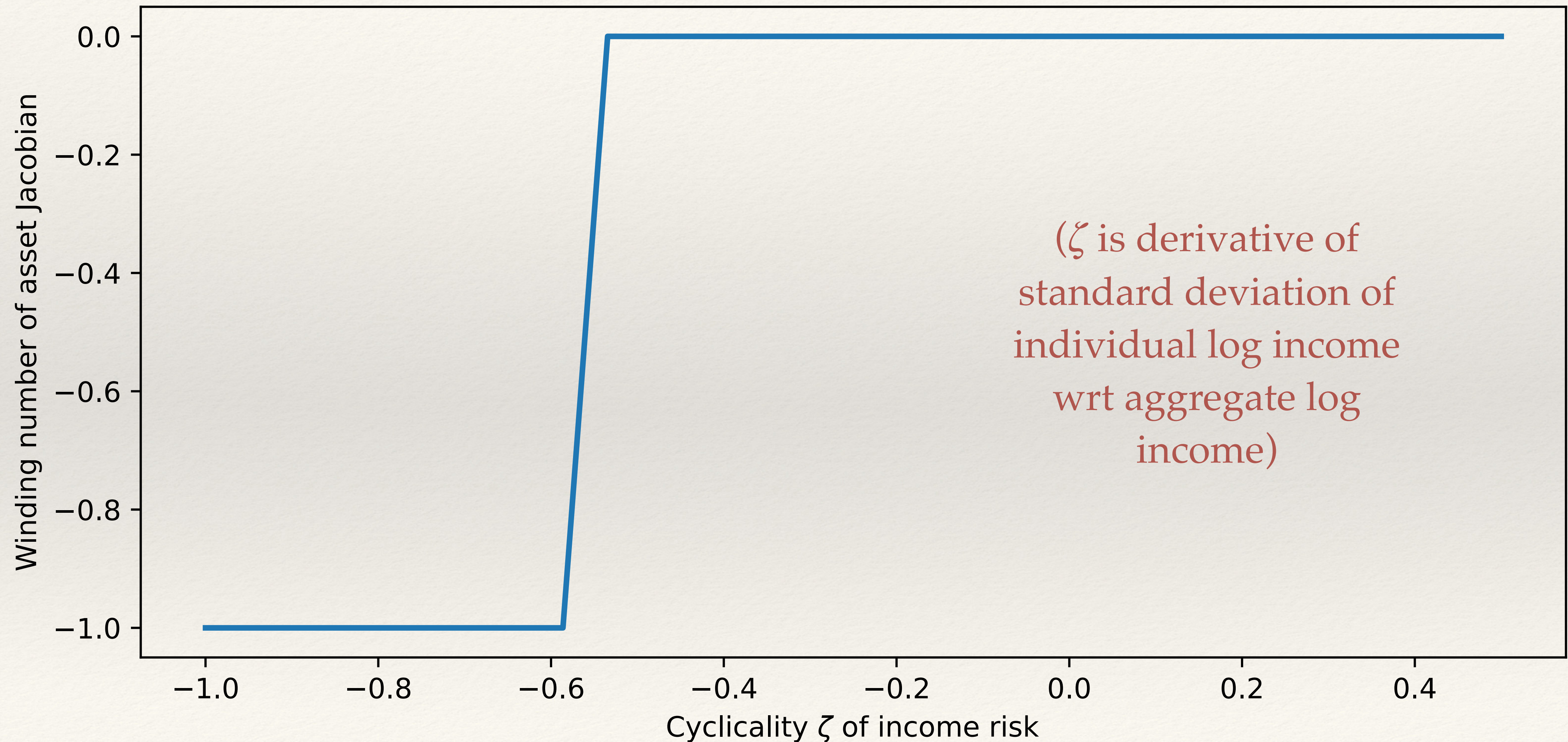


Get shape of indeterminacy from approx *SVD* null vector



Self-fulfilling boom
that is **very** long-
lasting: boom
increases income and
decreases future risk,
leading to self-
fulfilling boom in
consumption

How does winding number vary with cyclicality ζ of income risk?



Under the hood: winding number code

```
def sample_values_simple(j, K=8192):
    """Evaluate Laurent polynomial j(z) (with equally many
    positive and negative powers) counterclockwise at K evenly
    spaced roots of unity z, wrapping back around to z=1"""
    # j is (2*T-1,) array of Laurent coeffs (powers from -(T-1) to T-1)
    assert (K % 2 == 0) and (len(j) % 2 == 1) and (K > len(j))
    T = len(j) // 2 + 1

    # center j(z) at K/2, fill rest with zeros
    jhat = np.zeros(K)
    jhat[(K//2 - (T - 1)):(K//2 + T)] = j

    # take FFT of jhat to evaluate z^(K/2)*j(z) at clockwise roots of unity
    x = np.fft.fft(jhat)

    # divide by z^(K/2), which is alternating 1, -1 at roots of unity
    alt = np.tile([1, -1], K//2)
    x *= alt

    # return wrapped back to z=1, reversed to make counterclockwise
    return np.concatenate((x, [x[0]]))[:,::-1]
```

Once you have $j(z)$, it's easy to evaluate at sample points around unit circle very efficiently with the FFT—see left.

(Much better than a naive implementation.)

Then we just need to count the number of times the path wraps around the origin—tedious function to write, header below, full function in **winding_number.py**.

```
@jit
def winding_number_of_path(x, y):
    """Compute winding number around origin of (x,y) coordinates that make
    closed path by
    counting number of counterclockwise crossings of ray from (0,0) ->
    (infy,0) on x axis"""
```

Block quasi-Toeplitz case

- ❖ Say we have N^2 quasi-Toeplitz matrices from N unknowns to one of N targets
- ❖ Can think of this as being one **block quasi-Toeplitz operator**, like a quasi-Toeplitz but where entries are each $N \times N$ blocks
- ❖ Then $\{j_k\}_{k=-\infty}^{\infty}$ is two-sided sequence of $N \times N$ matrices, so *matrix-valued* $j(z)$:

$$j(z) \equiv \sum_{k=-\infty}^{\infty} j_k z^k$$

- ❖ Winding number test still holds generically, now for $\text{wind}(\det j)$
- ❖ Important case in practice

Beyond determinacy:
quasi-Toeplitz as a computational tool

Solving het-agent models to first order

- ❖ Two key considerations:
 - ❖ Size of idiosyncratic state space S
 - ❖ Number of endogenous aggregate variables N
- ❖ **State-space approach:** costly when S large. Has determinacy criterion.
[Reiter, Ahn-Kaplan-Moll-Winberry-Wolf, Bayer-Luetticke, ...]
- ❖ **Sequence-space approach:** fast when S large, costly when N large.
[Boppart-Krusell-Mitman, Auclert-Bardoczy-Rognlie-Straub, ...]
- ❖ How do we solve models when both S and N are large?

Further exploiting quasi-Toeplitz structure

- ❖ **Three ways of exploiting quasi-Toeplitz structure:**
 - ❖ **Winding number criterion on $\mathbf{j}$ for determinacy & existence [did this!]**
 - ❖ **“Truncation-free” solution working directly with $\mathbf{j}$ and $\mathbf{E}$ [next!]**
 - ❖ **Using $T(\mathbf{j}^{-1})$ as guess for $\mathbf{J}^{-1}$ gives rapid iterative solution, even when N large [next!]**

Operations with quasi-Toeplitz operators:
No more truncation!

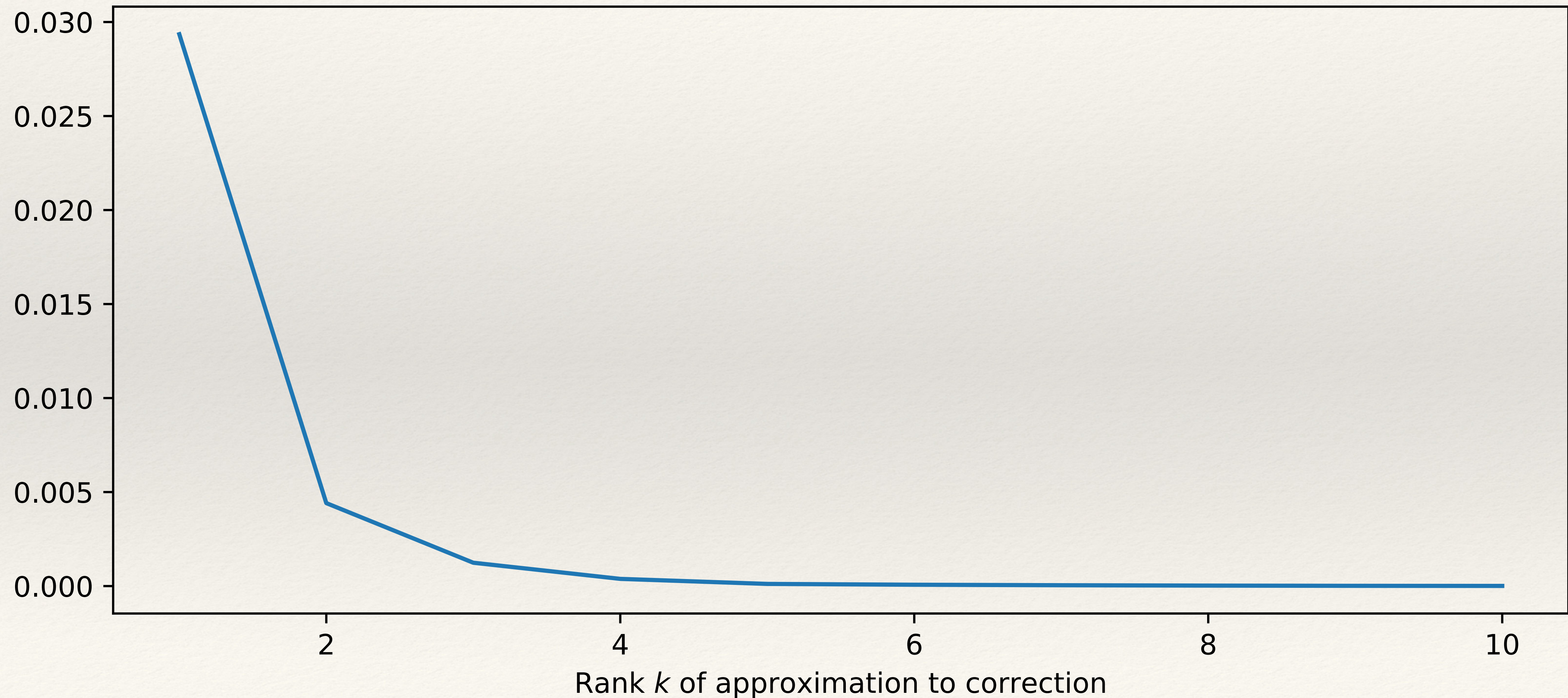
Directly use quasi-Toeplitz form

- ❖ We have quasi-Toeplitz representation $\mathbf{J} = T(\mathbf{j}) + \mathbf{E}$ of Jacobians
- ❖ So far, we've used the winding number of $\mathbf{j}$ to assess determinacy
- ❖ Going beyond this: directly do computations with this representation!
 - ❖ Benefit: $\mathbf{E}$ decays quickly to zero, and often close to low-rank $\mathbf{E} \approx \mathbf{U}\mathbf{V}'$
 - ❖ Cheap to get $T(\mathbf{j}_1\mathbf{j}_2)$, multiply $T(\mathbf{j}_1)\mathbf{U}'_2$, etc., to construct $\mathbf{J}_1\mathbf{J}_2$ (using FFT)
 - ❖ Similar, though a bit more complex, for inversion
- ❖ Working directly with quasi-Toeplitz (infinite!), not truncated matrices, avoids errors from truncation

[Bini, Masei, Robol 2019]

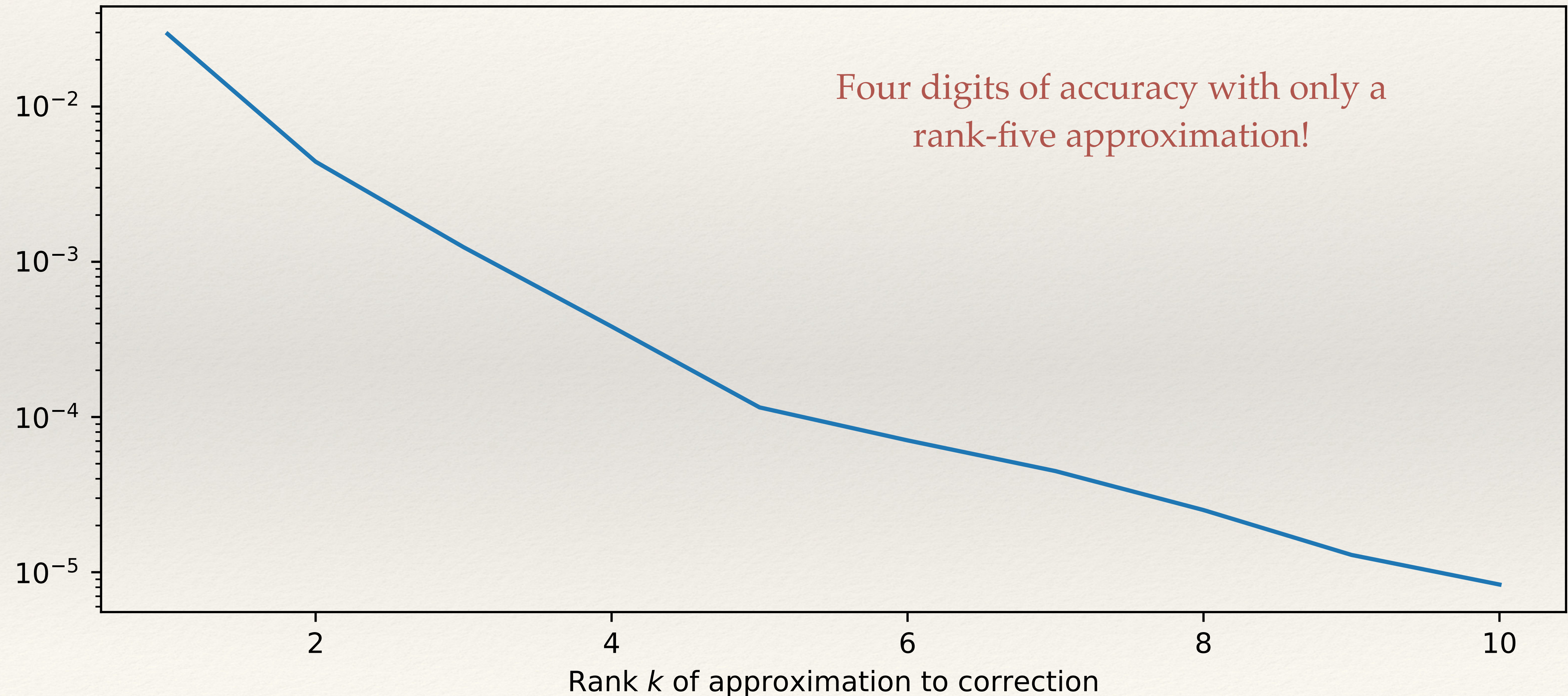
How well can we approximate $\mathbf{A}$?

Maximum error in $\mathbf{A}$ by approximation rank



Bit easier to visualize with a log scale...

Maximum error in **A** by approximation rank



Alternative: use structure for iterative solutions
(and solve giant models in the process!)

First point: easy to get Toeplitz part of inverse

- ❖ Suppose we want to solve $\mathbf{A}d\mathbf{Z} = d\mathbf{B}$
- ❖ $\mathbf{A}^{-1}$ is quasi-Toeplitz of form $T(\mathbf{a}^{-1}) + \mathbf{E}$, with $\mathbf{E}$ low-rank like we saw
- ❖ Key point: $\mathbf{a}^{-1}$ is **really** easy to calculate!
 - ❖ Get $a(z)$ at many z using FFT, then go from $a(z)^{-1}$ to $\mathbf{a}^{-1}$ with inverse FFT
 - ❖ Cost is only $O(T \log T)$, way cheaper than $O(T^3)$ matrix inversion
 - ❖ What can we do with just $T(\mathbf{a}^{-1})$?
 - ❖ [conceptually, $\mathbf{a}^{-1}$ is inverse for infinitely-well-anticipated shocks]

What can we do with $\mathbf{a}^{-1}$?

- ❖ Start with $(T(\mathbf{a}) + \mathbf{E})d\mathbf{Z} = d\mathbf{B}$, multiply both sides by $T(\mathbf{a}^{-1})$:

$$T(\mathbf{a}^{-1})(T(\mathbf{a}) + \mathbf{E})d\mathbf{Z} = T(\mathbf{a}^{-1})d\mathbf{B}$$

- ❖ Both $T(\mathbf{a}^{-1})T(\mathbf{a}) - \mathbf{I}$ and $T(\mathbf{a}^{-1})\mathbf{E}$ compact, well-approximated by low rank, so can be written in form

$$(\mathbf{I} + \mathbf{C})d\mathbf{Z} = d\mathbf{y}$$

- ❖ Iterative method (GMRES) **very** good at solving $(\mathbf{I} + \mathbf{C})^{-1}d\mathbf{y}$ if $\mathbf{C}$ low-rank

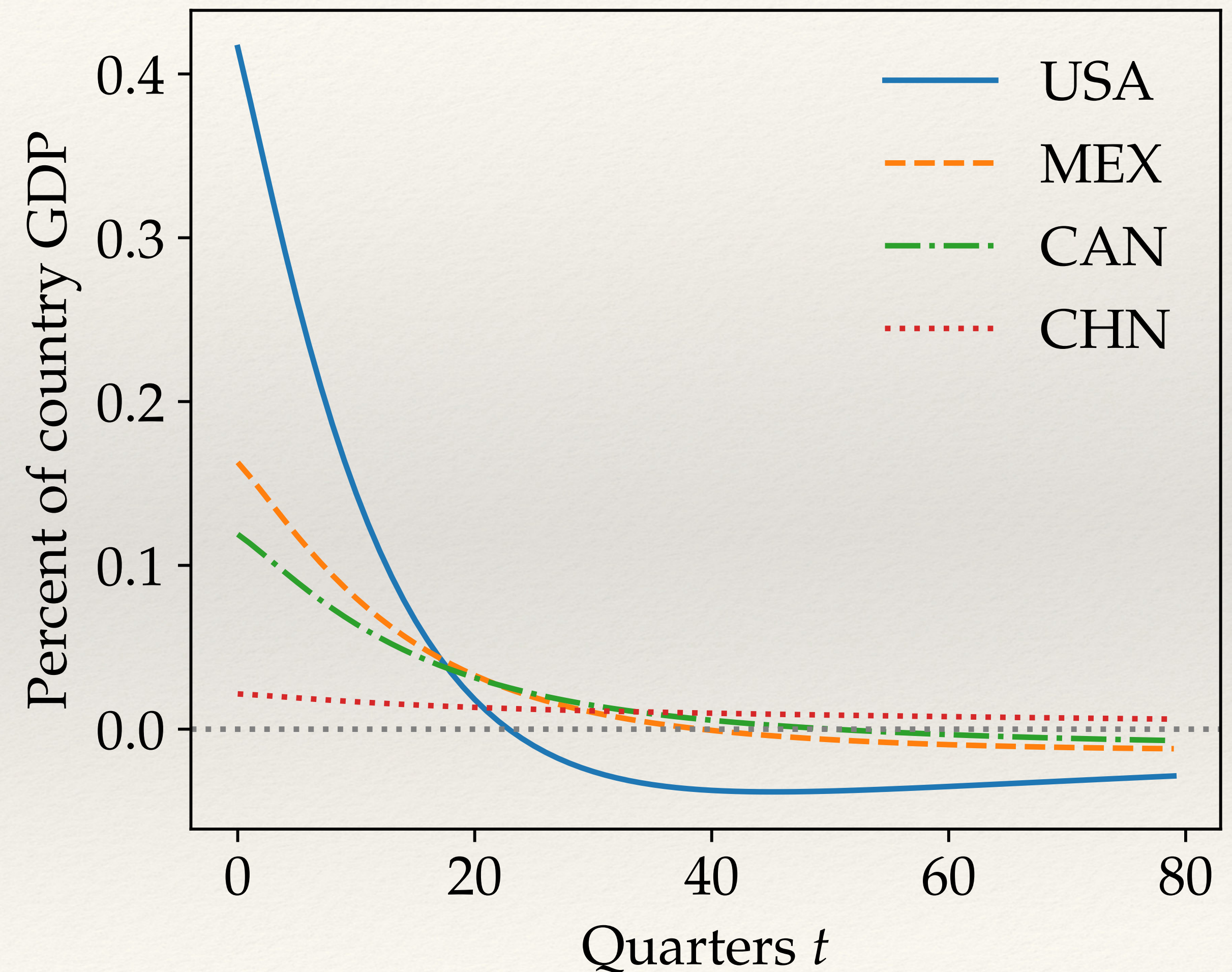
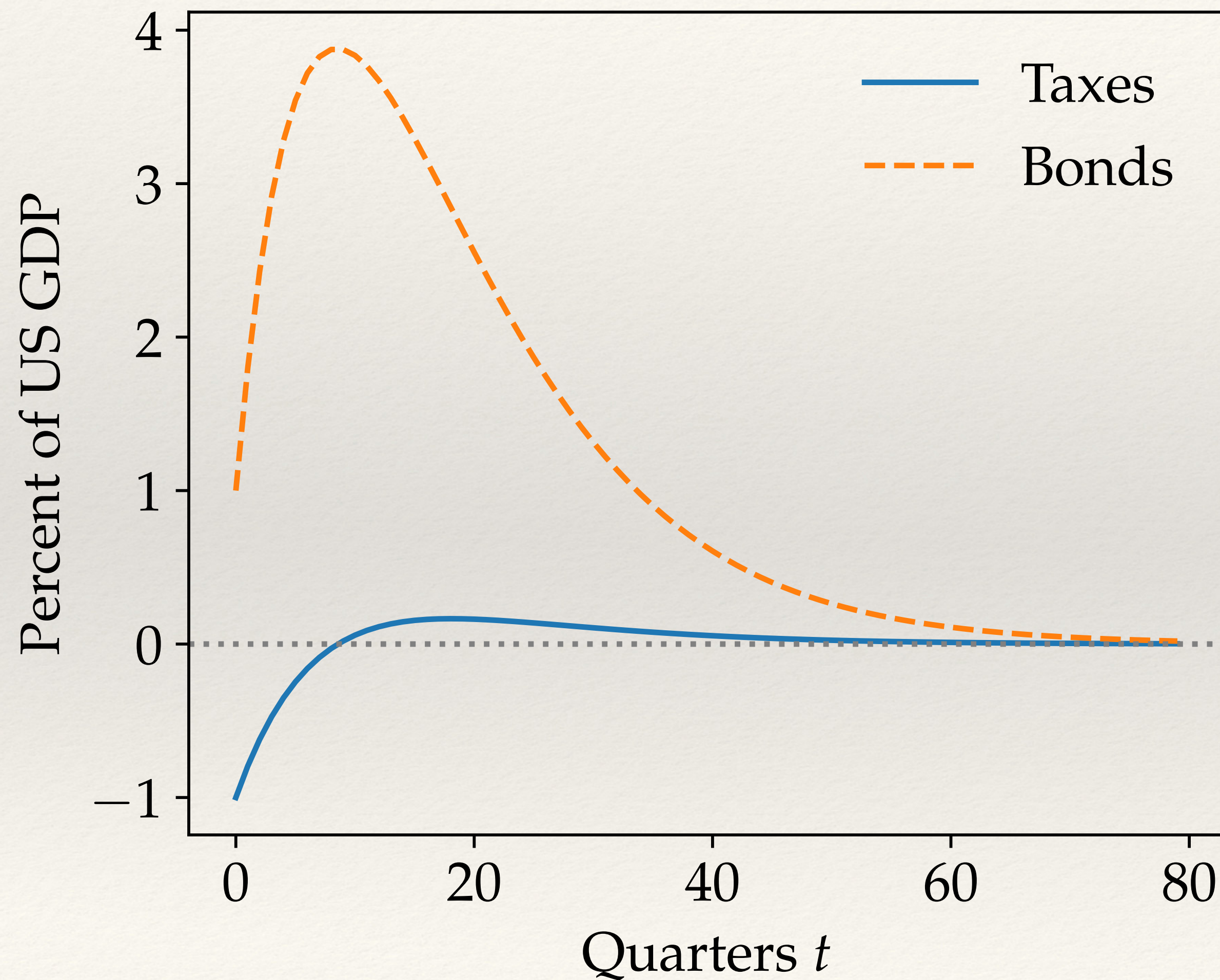
[Multiplying by $T(\mathbf{a}^{-1})$ is called “preconditioning”.]

- ❖ Cheap, doesn't require explicitly forming new matrices like $\mathbf{C}$

We expand this to HUGE model

- ❖ N -country extension of IKC model, constant r in each country n
- ❖ Fiscal policy in n chooses $\{B_t^n, T_t^n\}$ consistent with budget constraint
- ❖ n spends share $\Pi_{n,n'}$ on output from others n' , take from data for 190 countries
- ❖ Assume same HA model in each n , for simplicity assume all share $\mathbf{A}, \mathbf{M}$
- ❖ Solve for GDP $\{Y_{nt}\}$ in all N countries, in response to US deficit-financed tax cut, need long horizon $T = 1000$
- ❖ Usual sequence-space approach: Jacobian size $(190,000)^2$: can't even store!
- ❖ With iterative approach, solves in a few seconds on laptop!

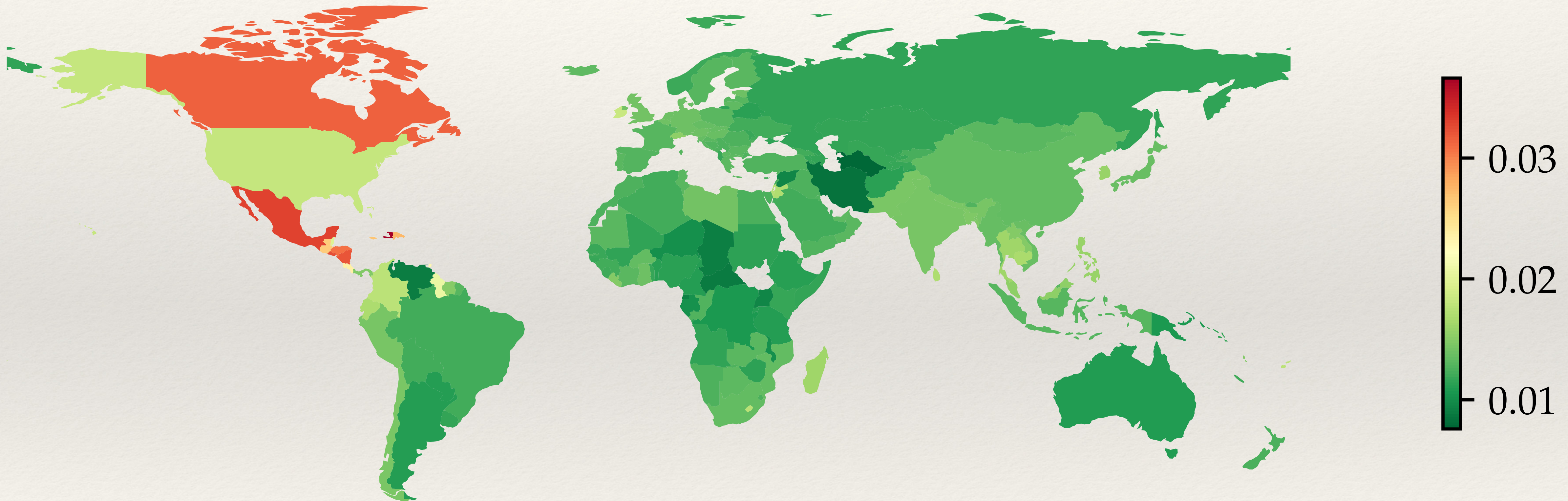
Peek at solution: selected countries over time



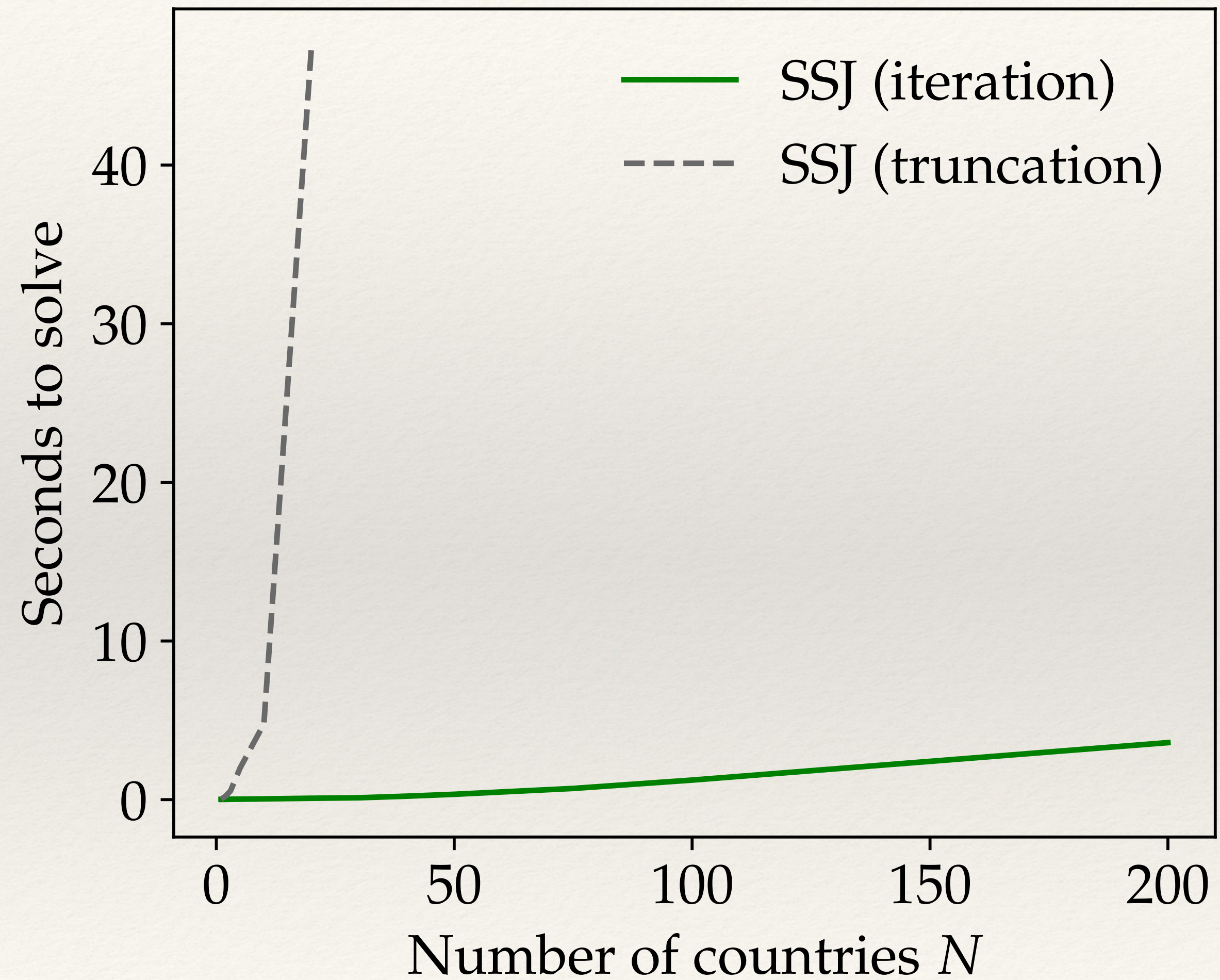
Peek at solution: on impact across countries



Peek at solution: after 20 quarters across countries

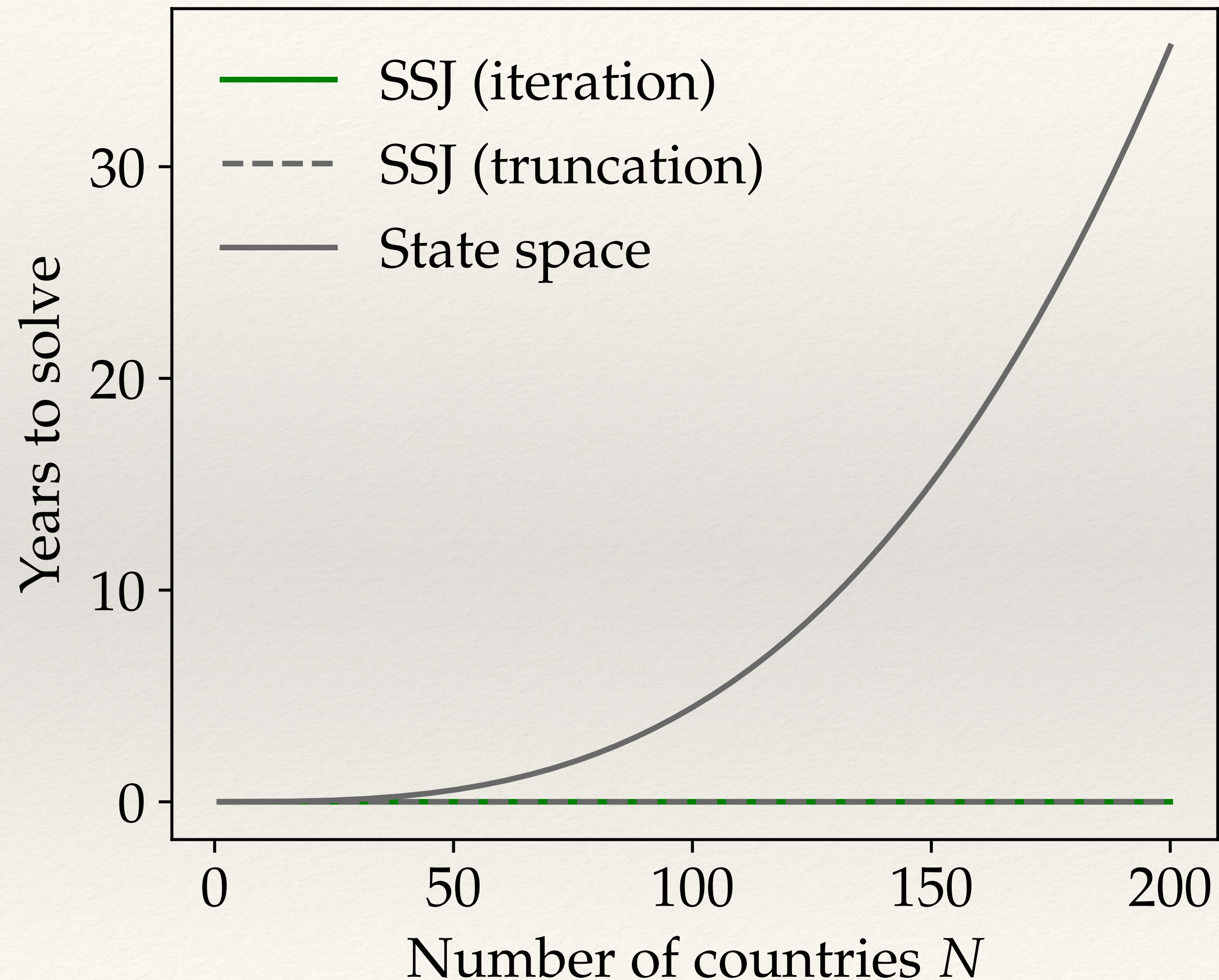


How fast is this?



How fast is this? Compare to state space

(Extrapolation
using cubic
cost of solving
state-space
system!)



Conclusion

- ❖ Quasi-Toeplitz structure of Jacobians delivers:
 - ❖ **winding number test** for determinacy
 - ❖ **truncation-free computations** exploiting the structure
 - ❖ **extremely fast iterative computations, even in huge models**
 - ❖ solves 190-country HANK in 3 seconds!!